



Waterproof Galaxy S4

Samsung launched a waterproof variant of Galaxy S4 dubbed as Galaxy S4 active. Yes, just like Xperia Z. <http://dgit.in/11uu5d2>



Wordpress woes

According to a recent study 50 most popular Wordpress plugins are vulnerable to common web attacks

by the relative importance of the adjacent sentences, similar to how PageRank calculates the relative importance of a web page. A similarity matrix is then constructed using this LexRank data, and all sentences with a score below a threshold value are rejected.

Speech recognition – Apple's Siri and Google's voice search

While the process of recognising spoken word and transcribing doesn't exactly fall under NLP, training computer systems to respond to the transcribed text is a very exciting application. The process of extracting meaningful and unambiguous text from an audio clip is done using an algorithm similar to the Hidden Markov Models described before. The audio clip is sent through a Fourier Transform, which splits up the stream into different frequencies (similar to an equalizer), and a correlation algorithm forms an association between the shape of the curve and the (known) word being spoken. A corpus of such training data is formed, which allows the system to assign probabilities to each word spoken by the user. Based on these probabilities, a Markov chain is constructed which gives a reasonable transcription of the spoken word into text.

Once the text has been extracted, virtual assistants like Siri, Google Voice Search and Siri-like alternatives on Android such as Iris, Vlingo and Robin parse the text and break it into parts of speech. The second part of the task involves extracting useful information from the text, such as the nature of the query and the best way to respond to it. These systems use a pre-indexed database that is used for replying to queries, which can include both offline data as well as online search engines to deliver the most relevant response.

Another age-old problem that is seeing a new lease of life is that of machine translation. Google Translate is a great example of applying the modern techniques of machine learning to automatic translations, as earlier attempts based on hard-coded rules failed while translating beyond simple phrases. This



Google debuted its updated Voice Search in Android 4.1 Jelly Bean

is still an open problem, however, as translating large chunks of texts (such as scientific papers, journals or novels) is far from perfect.

Not to be dissuaded however, researchers at Carnegie Mellon University have come up with a unique crowd-sourcing model of translating documents on the Internet using an app called Duolingo. The website (along with mobile apps for iOS and Android) gamifies the process of learning a new language, where users are rewarded with points for completing new lessons. The service, which is completely free of cost, tests these newly acquired language skills of its users by asking them to translate documents from/to the language they are learning. A statistical model is employed that correlates the translated documents submitted by thousands of its users to weed out any mistranslations.

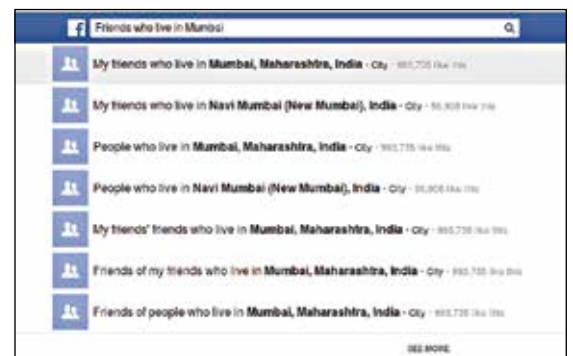
Facebook's newest feature, Graph Search employs Markov Models to break down the search string into a hierarchy of components that are then used as commands to query the database. For e.g., "Friends who live in Mumbai" will look for a union of all users who satisfy both conditions simultaneously. Permutations of this string such as "Friends in Mumbai" or "My Mumbai friends" also point to the same dataset, as the parsing engine is

intelligent enough to distinguish between the words in the string that are essential (friends, Mumbai) versus those that are non-essential (living, my). This allows for highly specific searches using data your friends have entered.

As statistical models for processing natural language data get more effective, the divide between programming languages and natural languages seems to be closing. Data scientists are finding new ways to parse human languages into tokens of data, leading to new high level programming languages being developed that lie very close to spoken languages on the scale, making programming accessible to a wider audience. Intelligent voice-activated assistants like Siri are indicative



Apple's Siri employs highly sophisticated classifiers to parse speech input



Facebook's Graph Search uses NLP to find the most relevant search results

of the massive progress that NLP has made over the years, but localising these systems into multiple languages, each with their own idiosyncrasies is another challenge that engineers are working hard to tackle. 